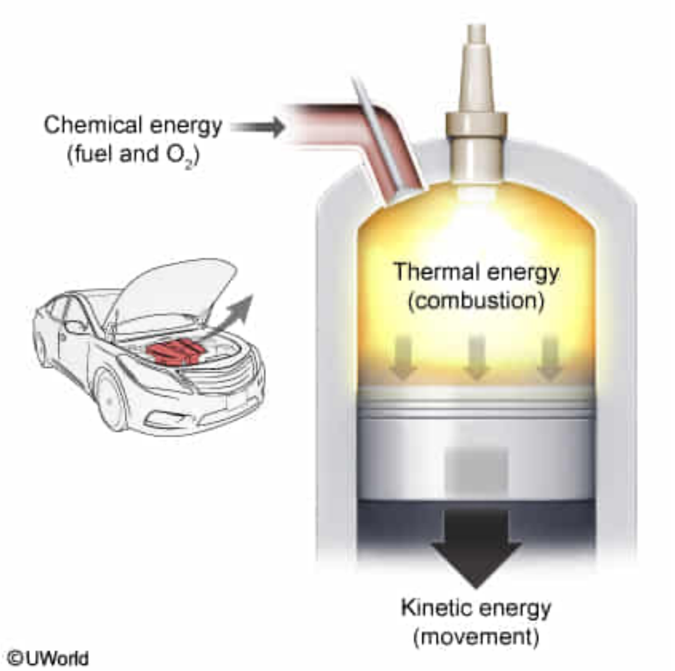
In which of the following processes does energy convert from chemical energy to thermal and kinetic energy?

- A. Digesting a piece of bread [15%]
- B. Toasting a piece of bread [4%]
- ✔ C. Running a combustion engine [74%]
- D. Charging a cell phone with an external battery [5%]

Correct
74% answered correctly

Explanation:



According to the law of **conservation of energy**, energy cannot be created or destroyed, only transformed from one form to another. Combustion is a chemical process through which the **chemical energy** stored in the molecular bonds of the reactants is released by creating new bonds with less energy.

The energy released from combustion is transformed into several forms. The combustion event produces a flash of light and a loud sound. The temperature of the gases increases **(thermal energy)**, and the gases expand to move the engine's components **(kinetic energy)**. Therefore, the conversion from chemical energy to thermal and kinetic energy takes place during the operation of a combustion engine.

(Choice A) Digestion is a chemical reaction that converts complex substances into simpler molecules that can be absorbed into the bloodstream. The process releases heat (thermal energy), but no kinetic energy is produced. Therefore, chemical energy in the bread is converted into chemical and thermal energy.

(Choice B) An electric current (electric energy) is used to make the toaster become hot (thermal energy) and change the chemical composition of bread to toast (chemical energy). Therefore, chemical energy is produced, not consumed, and no kinetic energy is produced in the process.

(Choice D) The external battery converts chemical energy to electric energy, which is converted back into chemical energy stored in the phone's battery. Therefore, the conversion in this system is chemical energy to electric energy and back to chemical energy.

Educational objective:
The law of conservation of energy states that energy cannot be created or destroyed, only transformed from one form to another. In a combustion engine, chemical energy is converted to thermal and kinetic energy.

Time spent: 0 sec	**QID**:401318
Subject: Physics	**System:** Mechanics and Energy